

3 The mammalian retina is a tissue that detects light.

(a) Rod cells and bipolar neurones are cell types found in the retina.

(i) What is the name of the chemical formed when light is absorbed by the photopigment found in rod cells?

(1)

- ☐ **A** cis-retinal
- ☐ **B** phytochrome
- ☐ **C** rhodopsin
- ☐ **D** trans-retinal

(ii) Which row in the table shows the effects of light absorption on cells in the retina?

(1)

	Rod cell	Bipolar neurone
☐ A	hyperpolarised	hyperpolarised
☐ B	hyperpolarised	depolarised
☐ C	depolarised	hyperpolarised
☐ D	depolarised	depolarised

(b) An investigation studied the effects of light intensity on the mammalian retina.

Samples of retina were exposed to different light intensities and the mean potential difference of the bipolar neurones was recorded.

The results are shown in the table.

Light intensity / a.u.	Mean potential difference / mV
2	11
4	17
8	19
12	20
16	20
20	20

(i) Describe the effect of light intensity on the mean potential difference of bipolar neurones.

(2)

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(ii) Calculate the percentage change in the mean potential difference when the light intensity increases from 4 to 12 a.u.

(2)

Answer%

(iii) Give **two** reasons why some people might have objections to the use of mammalian retinas in this investigation.

(2)

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(c) Which statement describes the response of the muscles in the iris to increasing light intensity?

(1)

- ☐ **A** circular and radial muscles contract
- ☐ **B** circular and radial muscles relax
- ☐ **C** circular muscles contract and radial muscles relax
- ☐ **D** circular muscles relax and radial muscles contract

(Total for Question 3 = 9 marks)

